Basics of IP Addressing

$2^9 = 512$

$2^{10} = 1024 = 1K$ (Kilo)

$2^{20} = 1024 \times 1024 = 1M$ (Mega)

$2^{30} = 1024 \times 1024 \times 1024 = 1G$ (Giga)

$2^{40} = 1T$ (Tera)

$2^{50} = 1P$ (Peta)

$2^{60} = 1E$ (Exa)

$2^{70} = 1Z$ (Zetta)

$2^{80} = 1Y$ (Yotta)

1mbps = 10^6 bits/sec
1gbps = 10^9 bits/sec
1kbps = 10^3 bits/sec

1 Byte = 8 bits

1KB = 1024 Bytes

1MB = 1024 KB (Kilo Byte)

1GB = 1024 MB (Mega Byte)

1TB = 1024 GB (Giga Byte)

1PB = 1024 TB (Tera Byte)

1EB = 1024 PB (Peta Byte)

Bit → b
Byte → B

1 ZB = 1024 EB (Exa Byte)

1 YB = 1024 ZB (Zetta Byte)

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Range of 8-bit ⇒ 0 - 255 i.e. 0 to $2^8 - 1$

∴ Range of n-bit ⇒ $0 - 2^n - 1$

Combinations of 1-bit ⇒ 0
1

Combinations of 2-bit ⇒ 00
01
10
11
$(2^2 = 4)$

Combinations of 3-bit ⇒ □□ (2-bits)
000
001
010
011
100
101
110
111
$(2^3 = 8)$

1-bit ⇒ 2 parts

2-bits ⇒ 4 parts

⋮

K-bits ⇒ 2^K parts

3-bit

2	1

2^2 parts → 2^1 (size of each part)

4 parts 2 (size of each part)

n-bit

K	n-K

2^K parts → 2^{n-K} (size of each part)

Introduction to IP Addressing:

IPv4 Address = 32-bit

Total no. of IP Addresses = 2^{32}

= 4294967296

$2^{30} \cdot 2^2$

= 4G (IP Addresses)

Initially in 1980, IP address was divided into two fixed parts i.e.

NID = 8 bit & HID = 24-bit

(Network ID) (Host ID)

32-bit

8	24
NID	HID

2^8 Networks

$= 256$ Networks

2^{24} host/network

$= 2^4 \cdot 2^{20}$ host/network

$= 16$ M host/network

$16\text{ M} \approx 1.6$ cr host/network

$N_1 = 2^{24}$ IP
$N_2 = 2^{24}$ IP
$N_3 = 2^{24}$ IP
$N_4 = 2^{24}$ IP
⋮ ⋮ ⋮
$N_{256} = 2^{24}$ IP

2^8 parts $= 256$ parts

256 Networks

$2^8 * 2^{24}$

2^8 → No. of N/W

2^{24} → No. of IP address in one network

IANA → Internet Assigned Number Authority

Disadvantage: There are only 256 networks and even a small organization must buy 16M computers (HOST) to purchase one network

Solution → Class Full Addressing

COMPUTER NETWORKS

Class - A		Class - B		Class - C	
8-bit	24-bit	16-bit	16-bit	24-bit	8-bit
NID	HID	NID	HID	NID	HID
2^8 N/W $= 256$ N/W		2^{16} N/W $= 2^6 * 2^{10}$ N/W		2^{24} N/W $= 2^{20} \cdot 2^4$ N/W $= 16M$ N/W	2^8 Host/N/W $= 256$ Host/N/W

$= 64K$ N/W

2^{24} Host/N/W

$= 2^{20} \cdot 2^{4}$ Host/N/W

$= 16M$ Host/N/W

≈ 1.6 Cr Host/N/W

2^{16} Host/N/W

$= 64K$ Host/N/W

e.g. of class A → designed for Big Govt. org. like NASA

e.g. of class B → designed for MNC's like TCS, wipro etc.

e.g. of class C → designed for small org. like schools & colleges

Class - A	Class - B	Class - C
2^{24} IP Address/ Network	2^{16} IP Address/ Network	2^{8} IP Address/ Network

- Let X = 200 IP Address

Class C network will be assigned

Wastage $= 2^8 - 200$

$= 256 - 200$

$= 56$ IP Address

- Let $Y = 500$ IP Address

  class B network will be assigned

  Wastage $= 2^{16} - 500$

  $= 65536 - 500$

  $= 65036$ IP Address

- Let $Z = 70,000$ IP Address

  class A network will be assigned

  Wastage $= 2^{24} - 70,000$

  $= 16707126$ IP Address